

ACTION RESEARCH IN SOFTWARE ENGINEERING

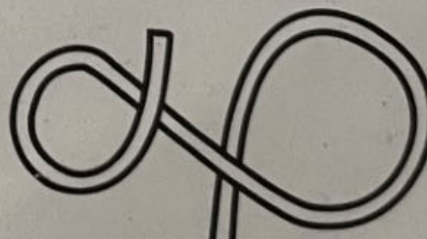
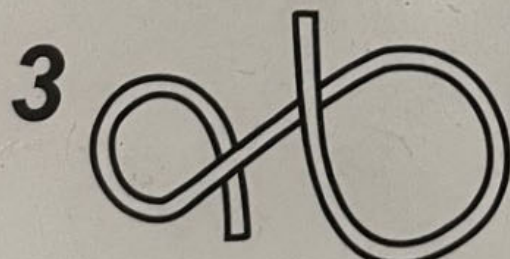
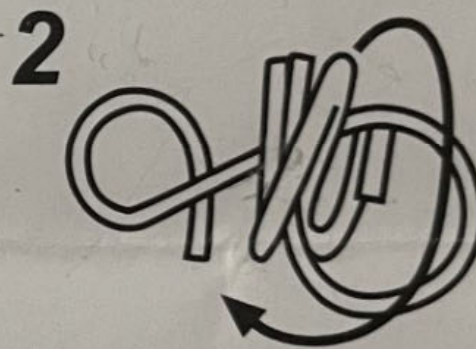
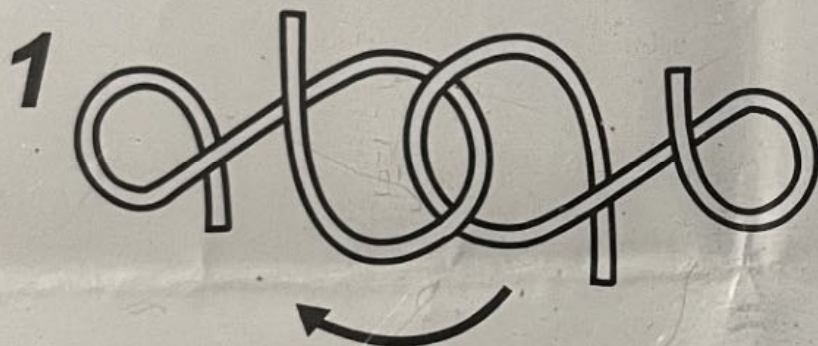
You must be the change you wish to see in the world.

- Mahatma Gandhi

MIROSLAW STARON
COMPUTER SCIENCE AND ENGINEERING, CHALMERS UNIVERSITY OF TECHNOLOGY AND UNIVERSITY OF GOTHENBURG

PUZZLE

PUZZLE
GENIUS RATING



Action Research in Software Engineering

Theory and Applications

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Outline of this lecture

- Action research as a methodology
 - Important characteristics
- Phases of action research
 - Crucial elements
- Threats to validity
 - What can go wrong
- Ethics in Action Research
 - What we can and cannot do
- Examples
 - Metrics research

Materials for the lecture today

- Ochodek, M., Hebig, R., Meding, W., Frost, G., & Staron, M. (2019). Recognizing lines of code violating company-specific coding guidelines using machine learning. *Empirical Software Engineering*, 1-46.
– <https://tinyurl.com/t5f7oux>
- Staron, M. (2019). *Action Research in Software Engineering: Theory and Applications*. Springer Nature.
– <https://tinyurl.com/uxhdttd8>
- Paul Ralph. 2021. ACM SIGSOFT Empirical Standards Released. *SIGSOFT Softw. Eng. Notes* 46, 1 (January 2021), 19.
– <https://doi.org/10.1145/3437479.3437483>
- Staron, Mirosław. "Guidelines for Conducting Action Research Studies in Software Engineering." *e-Informatica Software Engineering Journal* 19.1 (2025): 250105.
– Guidelines for Conducting Action Research Studies in Software Engineering
- Staron, Mirosław. "Teaching Action Research." *Handbook on Teaching Empirical Software Engineering*. Cham: Springer Nature Switzerland, 2024. 387-411.
– <https://arxiv.org/pdf/2408.02399>
- Dos Santos, Paulo Sergio Medeiros, and Guilherme Horta Travassos. "Action research can swing the balance in experimental software engineering." *Advances in computers*. Vol. 83. Elsevier, 2011. 205-276.

Continuing Prospects for an Engineering Discipline of Software

Mary Shaw, Carnegie Mellon University Institute for Software Research

Action Research Can Swing the Balance in Experimental Software Engineering

Paulo Sérgio Medeiros dos Santos

Guilherme Horta Travassos

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Why action research

- Industry criticizes academia for producing research results that is not applicable
- Industry wants the research results to be used in practice
 - Tools, methods, practices
- Academic software engineering research is based on rigorous research methodologies
 - Replicability of results, generalizability of conclusions
- Action research as a methodology helps to fulfill requirements from both academia and industry
 - You must do the work yourself, though



Contents lists available at ScienceDirect

Information and Software Technology

journal homepage: www.elsevier.com/locate/infsof



Guiding the selection of research methodology in industry–academia collaboration in software engineering

Claes Wohlin^{a,*}, Per Runeson^b

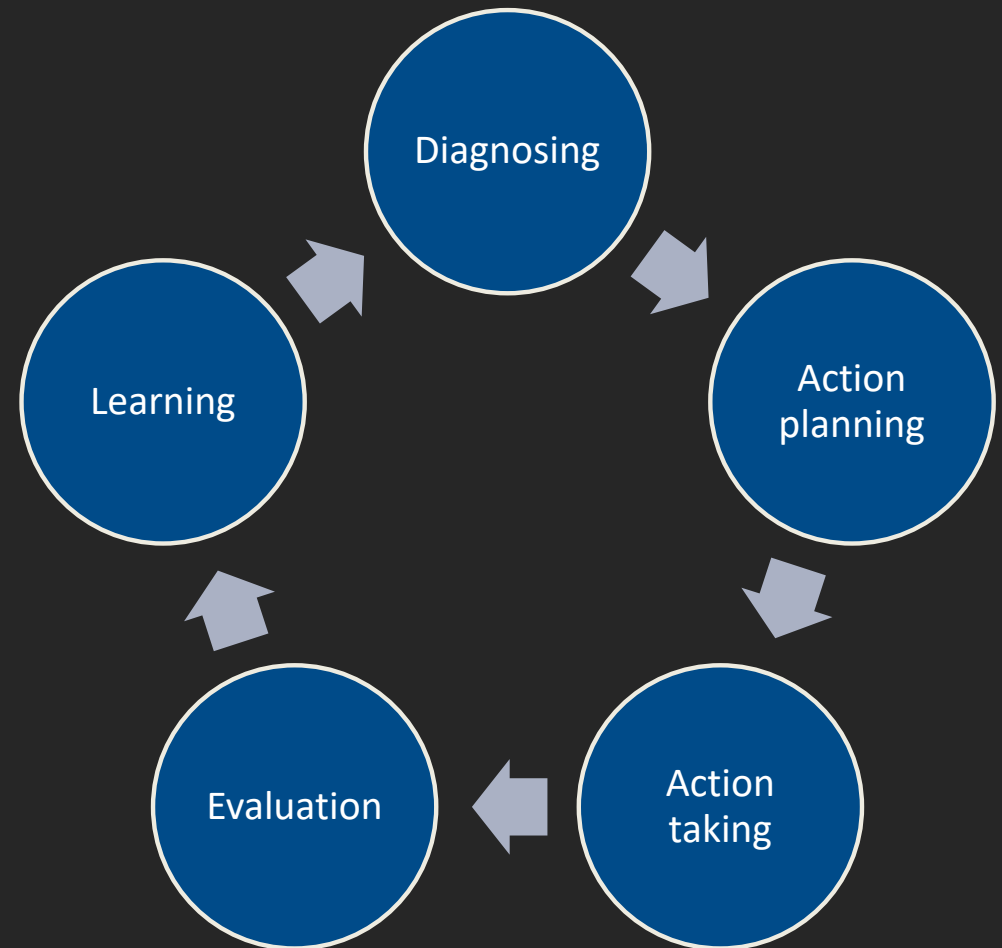
^a Blekinge Institute of Technology, Karlskrona, Sweden

^b Lund University, Lund, Sweden

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Action research

- Action research is a disciplined process of inquiry conducted by and for those taking the action
- The primary reason for engaging in action research is to assist the actor in improving and/or refining his or her actions





Important aspects of action research

- Action / Intervention
 - We do something at the company
- Observation / Measurement
 - We measure the effect of our action
- Context / Host
 - We need to act at the company's premises and intervene with the company's processes



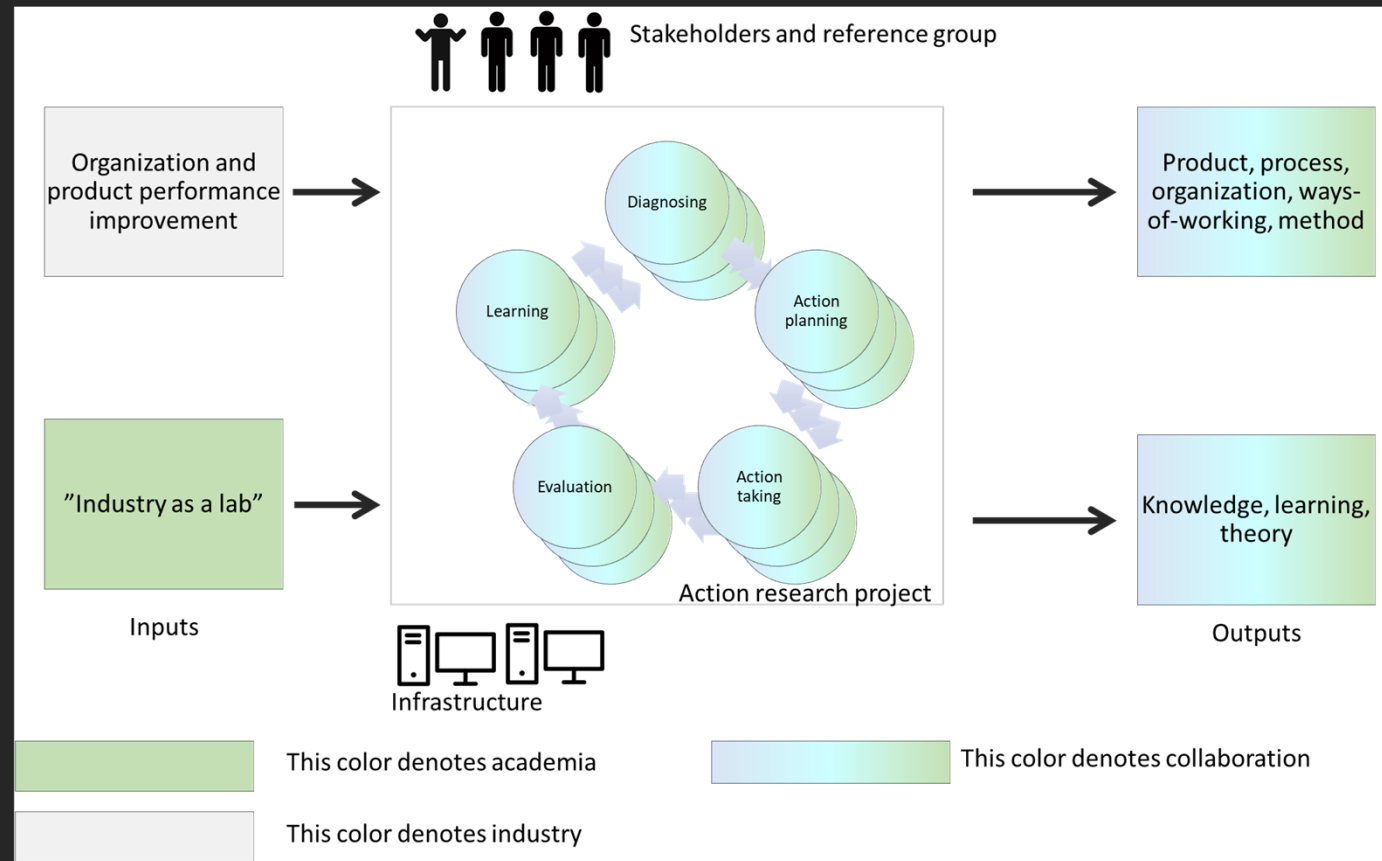
Specific Attributes

Importance	Attribute
Essential	☐ describes the context or site of the intervention(s) ☐ describes the intervention(s) in detail ☐ describes the relationship between the researcher and the host organization⁵ ☐ reports the length of the project and describes the longitudinal dimension of the research design ☐ describes the interactions between researcher(s) and host organization(s)—what the interventions were, who intervened and with which part of the organization, as well as the outcome of the interventions ☐ describes how interventions were determined (by management, researchers, or participative/co-determination process) ☐ explains research cycles or phases, if any, and their relationship to the intervention(s)⁶ ☐ explains how the interventions are evaluated⁷ ☐ reports participant or stakeholder reactions to interventions ☐ presents a clear and well-argued chain-of-evidence from observations to findings ☐ reports lessons learned by the organization ☐ researchers reflect on their own possible biases
Desirable	☐ uses direct quotations extensively ☐ uses member checking to assess resonance ☐ findings plausibly transferable to other contexts ☐ triangulation across quantitative and qualitative data
Extraordinary	☐ research team with triangulation across researchers (to mitigate researcher h



SE Action research

- Includes stakeholders
- Infrastructure
- Contracts
- IPR regulations



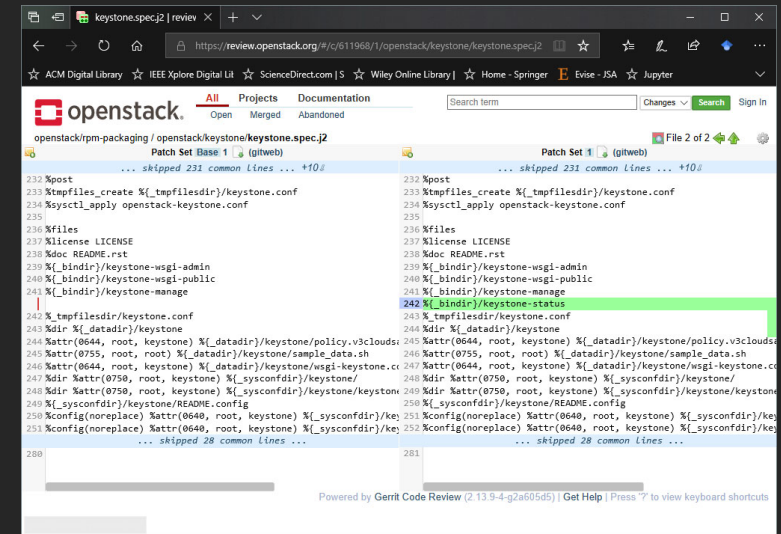
Host organization

- Action research is embedded in the host organization
- The research study involves practitioners
 - software engineers, project managers, architects, business analysts, deployment eng.
- Researchers in the project are treated as part of the host organization
 - Practitioners are part of the research team



Intervention or action

- What we do at the host organization
 - Change in the process, ways of working
 - New tool, method, technique
 - New insights that impact the host organization
- Important characteristics
 - has observable (preferably measurable) effect
 - is based on theory, describing the *what* and *why* we intervene

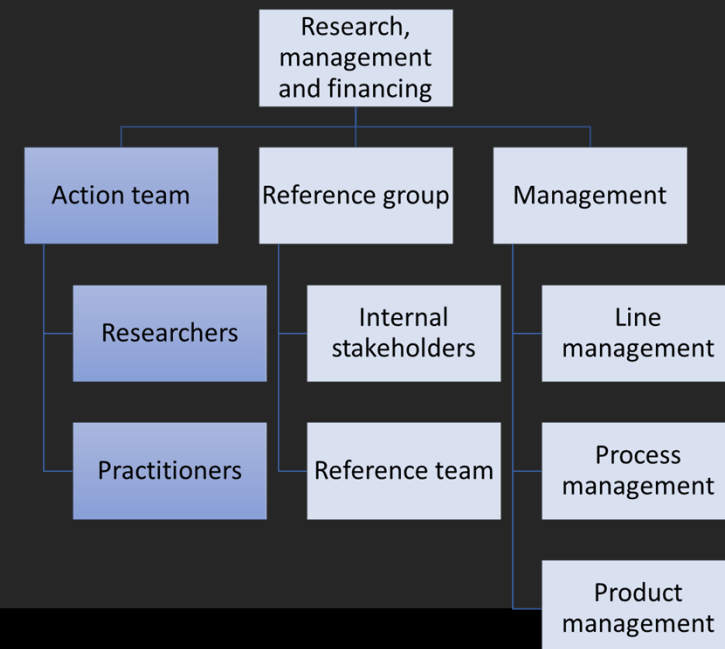
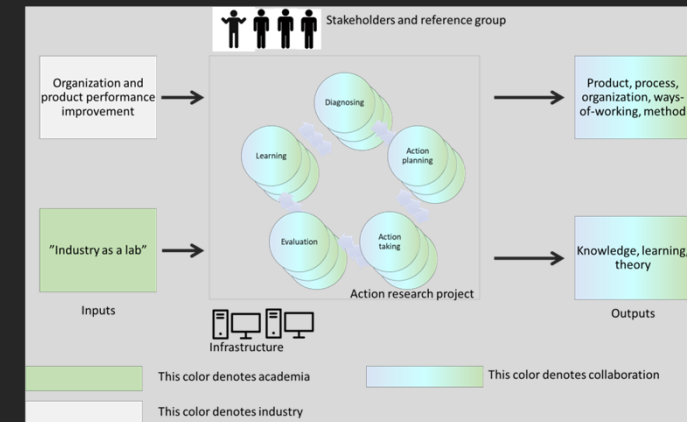


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BehaviorDescription_Special_signal_output_requirement	1	W	2015-04-16	admin	2015-01-23		
BehaviorDescription_Turbo_boost_pressure_presense	0	W					
BehaviorDescription_Immobilizer_engine_crank_audible	0	W					
BehaviorDescription_EcoCruise_EMS	9	W					
BehaviorDescription_Engine_starting_Aid	7	W					
BehaviorDescription_Factory_Mode	1	W					
BehaviorDescription_Coolant_Pump_Control	1	W					
BehaviorDescription_Air_Engine_speed_request	5	W					
BehaviorDescription_Active_Regeneration	0	W					
BehaviorDescription_Cross_Country_Engine_Speed...	3	W					
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Output_signals_in_non_operational_modes_and_d...	2	W					
Idle_shutdown_control	1	W					
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BehaviorDescription_TSCl_EngSpeed_arbitration	0	W					
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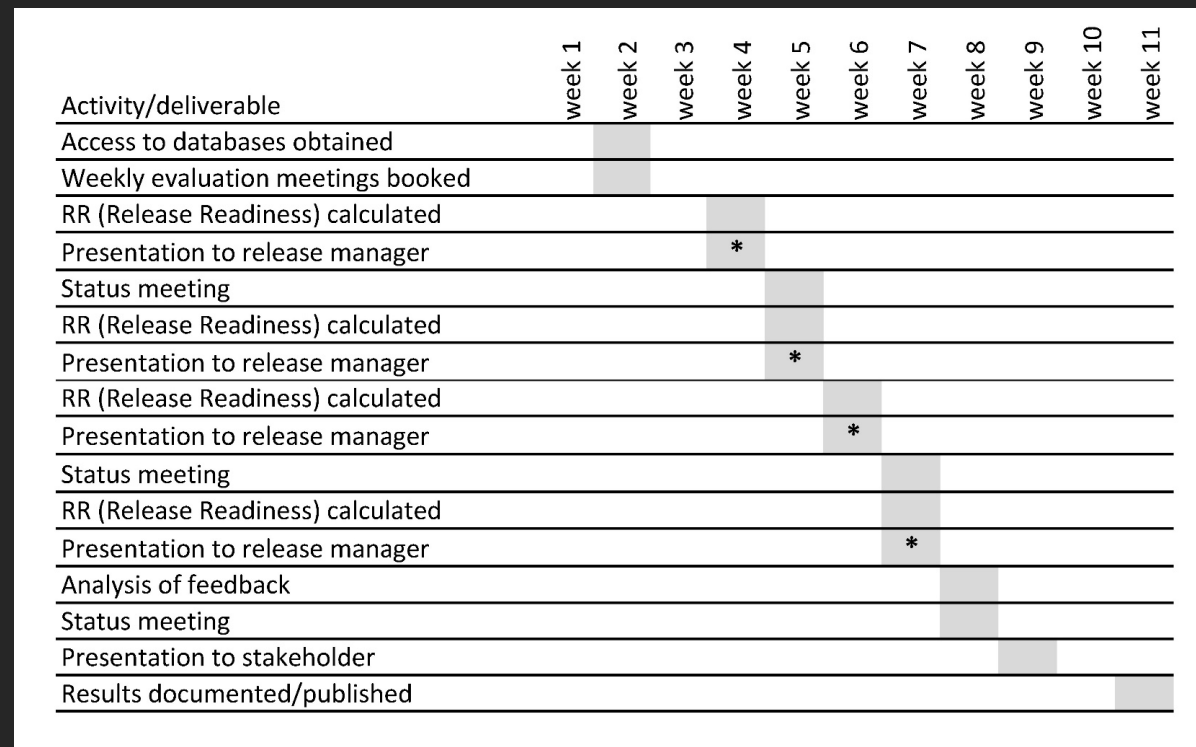
Diagnosing

- Purpose
 - Understand the nature of the problem – *What is the problem?*
- Main challenges
 - "Look under the hood" of the organization
 - Establish trust
- Major methods
 - Interviews
 - Observations



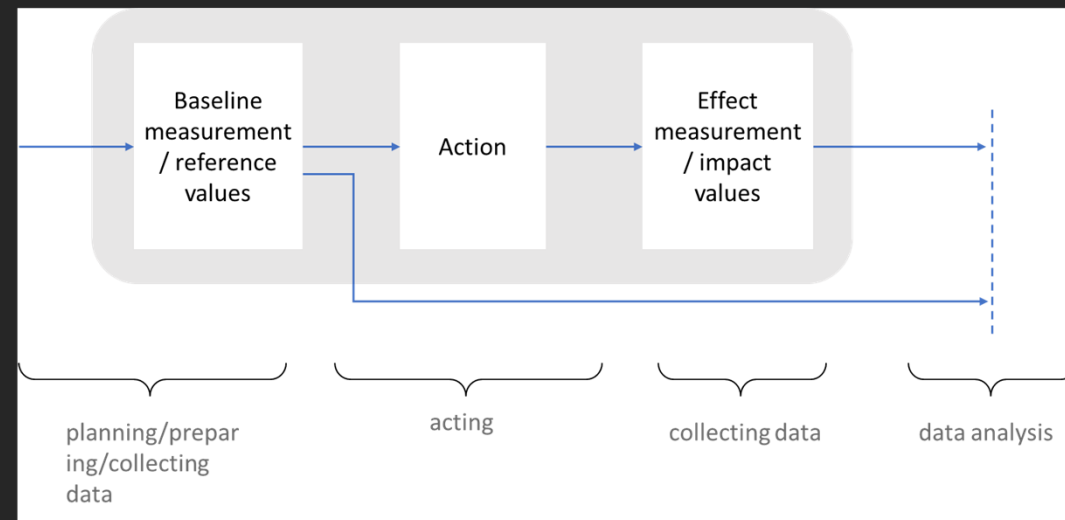
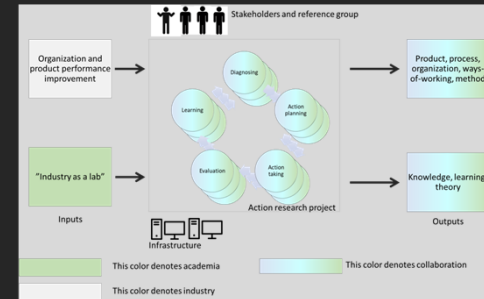
Action planning

- Purpose
 - Set up the project structure for each cycle
- Main challenges
 - Focus on deliverables
 - Combining impact with theory building
 - Planning for dissemination and feedback
- Major methods
 - Gantt charts
 - Workshops



Action taking

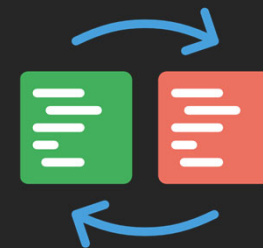
- Purpose
 - To intervene with the company's operations
- Main challenges
 - Data collection
 - Biases
 - Knowing when to pivot and adjust
- Major methods
 - Assessment frameworks (e.g. GQM)
 - SE tools and methods



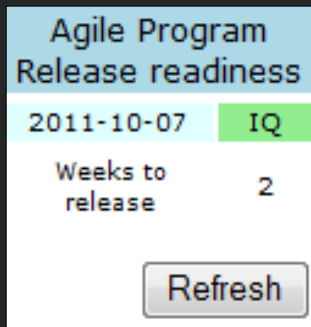
Intervention / Action



- Intervention is an activity that is done at the host organization, which has the goal to change the current practices at the organization
 - An action
- Examples
 - Introduction or removal of a tool
 - Introduction or removal of a checklist
 - Introduction or removal of a role
 - Change of an activity, role, tool
 - Workshops with the goal to change practices

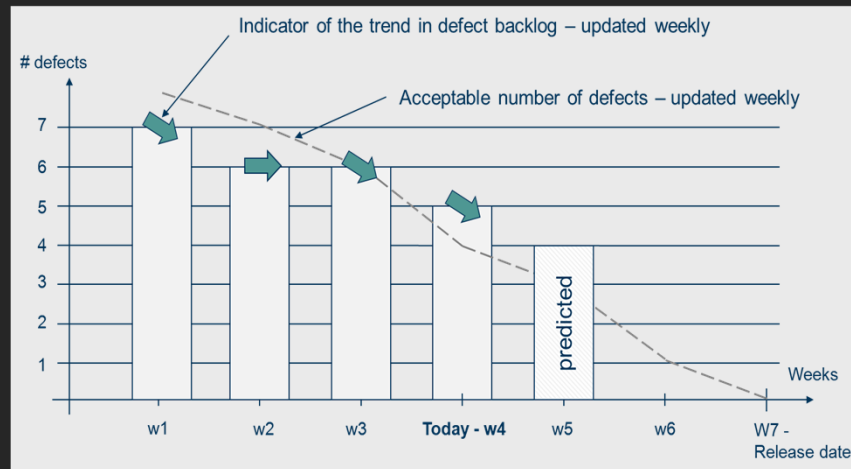


(My) interventions – three examples



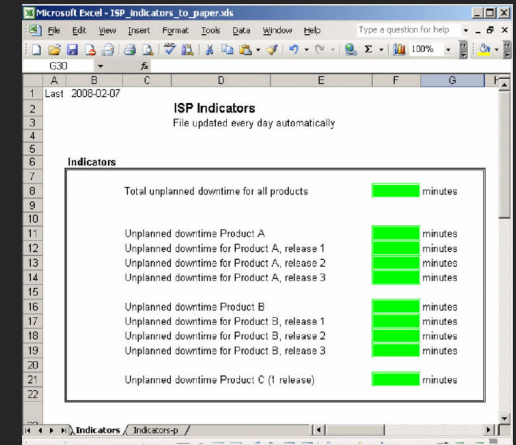
Indicator showing when the measured software product is ready to release.

Impact: increase delivery speed



Indicator to predict the number of found defects per week

Impact: increase software quality

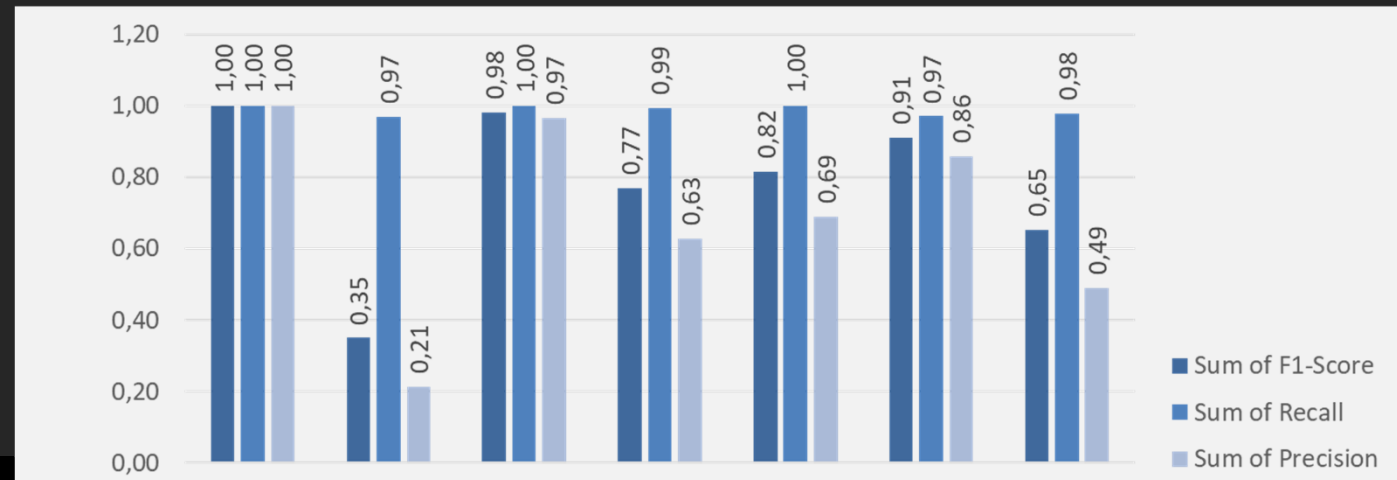
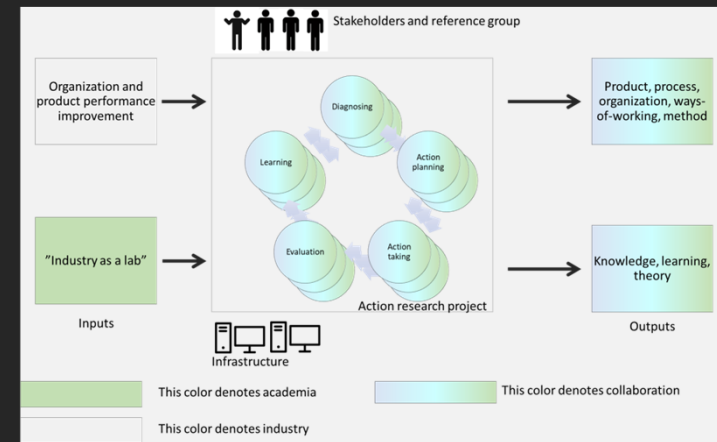


Measurement systems (40,000)

Impact: increased awareness in the organization

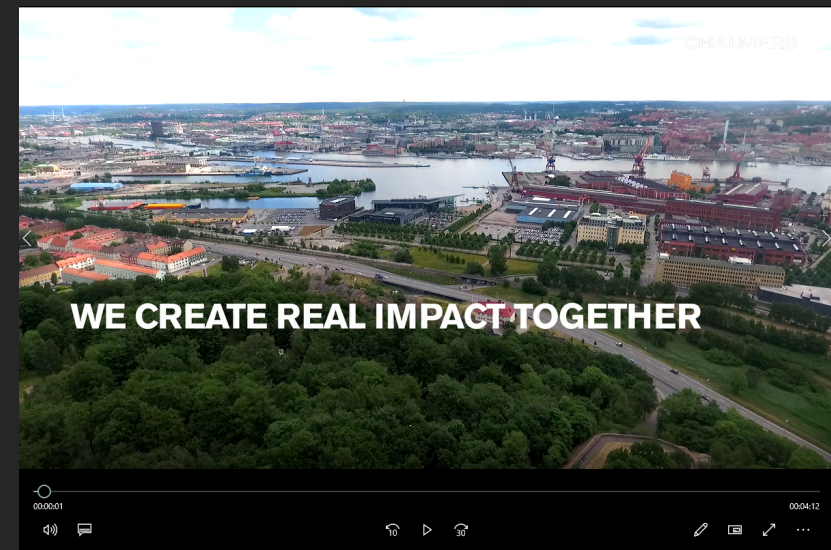
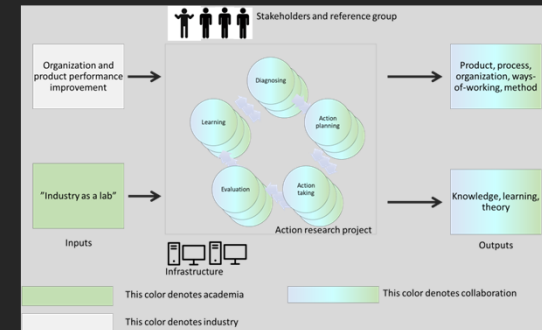
Evaluation

- Purpose
 - To understand the impact of the actions
- Main challenges
 - Data quality
 - Combining the quantitative and qualitative data
- Major methods
 - Statistics
 - Machine learning
 - Interviews
 - Workshops
 - Surveys



Specifying and documenting Learning

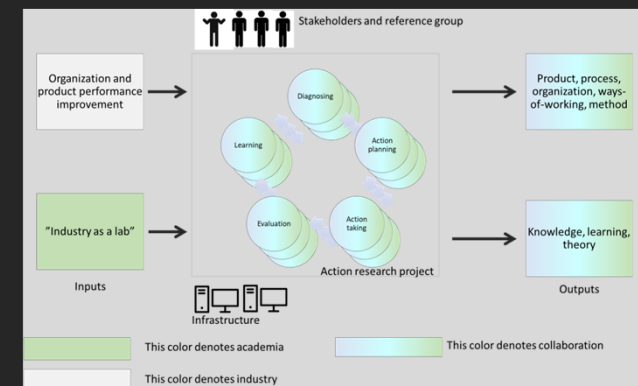
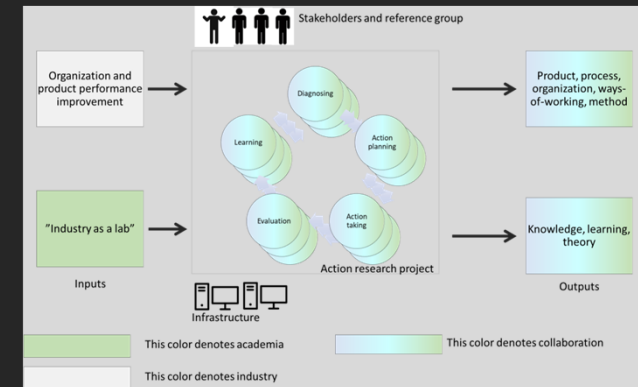
- Purpose
 - Document and disseminate good examples and practices
- Main challenges
 - NIH – "Not Invented Here" syndrome
 - Identifying key factors
- Major methods
 - Presentations
 - Videos
 - Podcasts



Action research is a key success factor

Company-to-company-to-academia research environment

- Working on site at a company
 - Initiates new ideas and research challenges
 - Enables direct validation of ideas and results
 - Reduces research cycle times
- Constant collaboration allows to reduce the effort for the companies
- IPR regulations/contracts enables co-development of research results



Context

Software development **Team A** of **Company A** which develops a web service for checking car's registration numbers. The team consists of software developers, testers and architects. All team members are involved in the research.

Goal

The team needs to **improve the quality of their product** by minimizing the number of times the product restarts. They need to **understand how to measure the downtime** of their product during the development to forecast the downtime at the customer's site. They need to find or develop a new measure to quantify downtime during development. This measurement needs to be representative for the downtime for the product in field, i.e. once the product is deployed at the customer.

Actors

Team A is the main actor in the project as they are working on the introduction of the measurement. The **role of the university researcher** is to review existing, documented and published experiences with downtime measurements. The **role of a quality manager and release manager** are to provide reference whether the measurement leads to the expected quality increase. The **role of the product manager** is to provide the evaluation

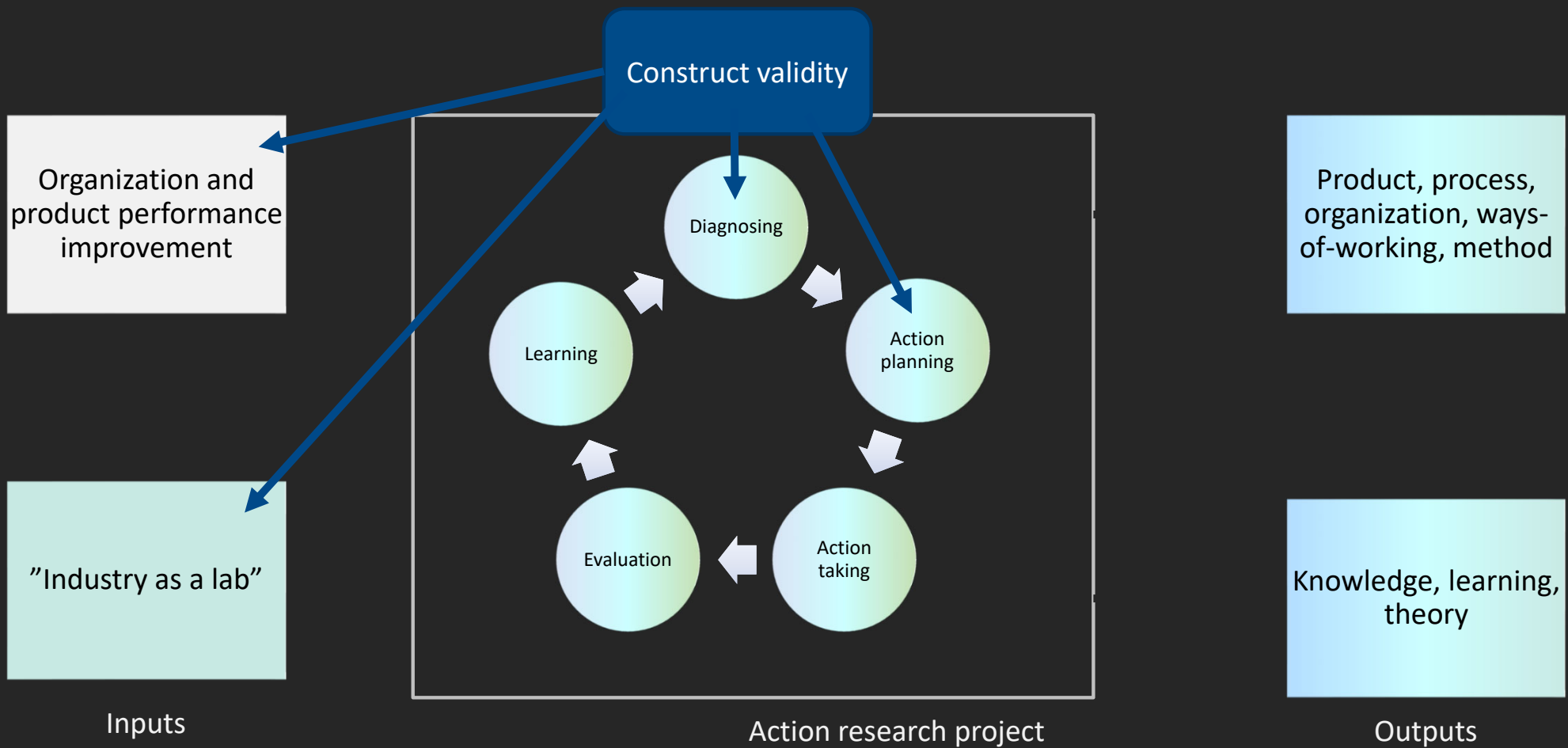
AR proposal

- Context
- Goal
- Actors
- Intended outcome
- Actions / Interventions
- Data collection and measurement
- Evaluation
- What we want to learn
- Ethical considerations



AR threats to validity

- Construct validity
 - How well we designed the study, e.g. Where we conduct the study
- Internal validity
 - How well we conducted the study, e.g. Which tools we used
- Conclusion validity
 - How well we understood the observations, e.g.
- External validity
 - How well we can generalize the results, e.g. Who is impacted



Threat	Explanation	Example
Inadequate Pre-operational Explication of Constructs.	This means that our constructs and measures are not related to the concepts which we want to evaluate. We plan to study one thing, but we measure something else.	We plan to study quality, but we measure software.
Mono-Operation Bias	This means that we study only one specific instance of the setting. This is quite common for action research projects as we often are embedded in the context.	We study only one development team in a time; instead we study several teams and thus vary the context there.
Mono-Method Bias	This means that we measure the effect with only one variable, thus do not capture the breadth of the entire effects.	We measure the impact only on the product team, forgetting to measure effects of the action on the product.
Interaction of Different Treatments	This means that the observed results of the study can be caused by other treatments, not the one intended. In action research this happens if our actions are taken in parallel with other improvement activities, which we do not control.	We introduced a new measure and observed a decrease of defects; at the same time the organization adopted a new testing method, which reduced the number of defects.
Interaction of Testing and Treatment	This means that just taking the action in the context makes the change; our preparations and diagnosing of the action may change the context of bias the subjects.	We ask about the product quality and therefore improve it; that the quality improved.

Construct validity

- How we designed the study based on the goals
- How we understood the empirical phenomena
- How we can measure what we need to observe
- How we can avoid confounding factors



Stakeholders and reference group



Action research project



Infrastructure

Action research – recap

- Focus on action and the host organization
- Impact on the organization and the theory
- Growing popularity in SE
- Great for a thesis in SE
 - If you have a host organization

Specific Attributes

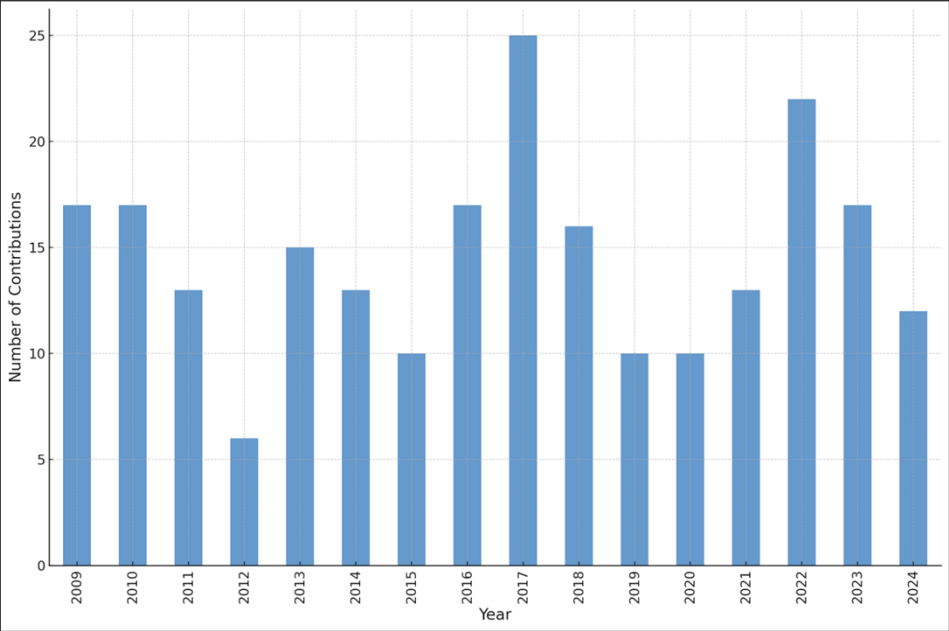
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Desirable	☐ uses direct quotations extensively ☐ uses member checking to assess resonance ☐ findings plausibly transferable to other contexts ☐ triangulation across quantitative and qualitative data
Extraordinary	☐ research team with triangulation across researchers (to mitigate researcher b



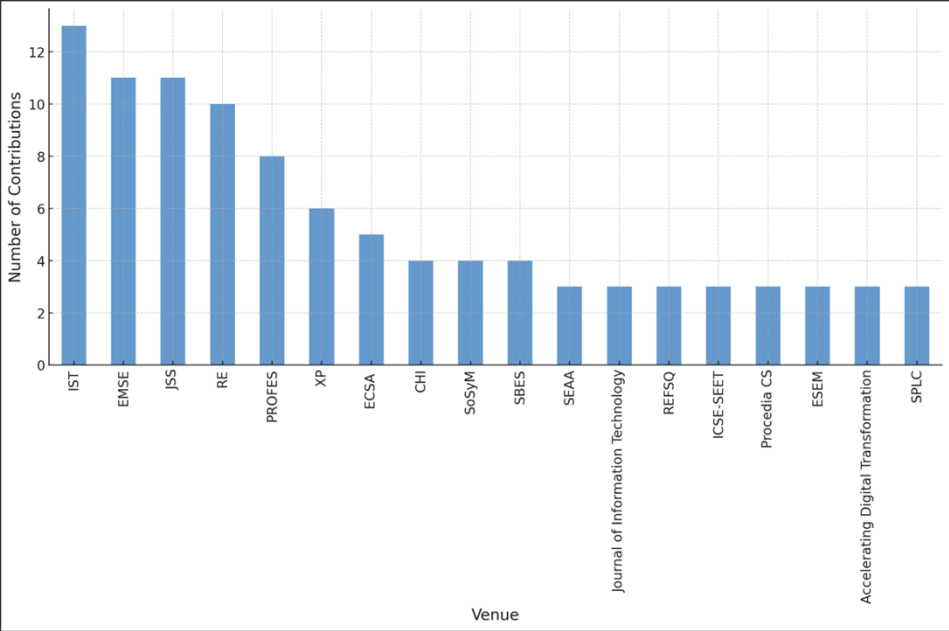
SYSTEMATIC REVIEW OF AR IN SE

Database	Search string	Date query executed	#Results	#Included
IEEE	"action research" + "software engineering"	2024-08-14	83	21
ScienceDirect	"action research" + "software engineering"	2024-08-14	423	34
ACM	"action research" + "software engineering"	2024-08-12	556	87
SpringerLink	"action research" + "software engineering"	2024-08-13	490	87
Scopus	"action research" + "software engineering"	2024-08-20	141	8

Table 1. Sources of papers included in the systematic review.



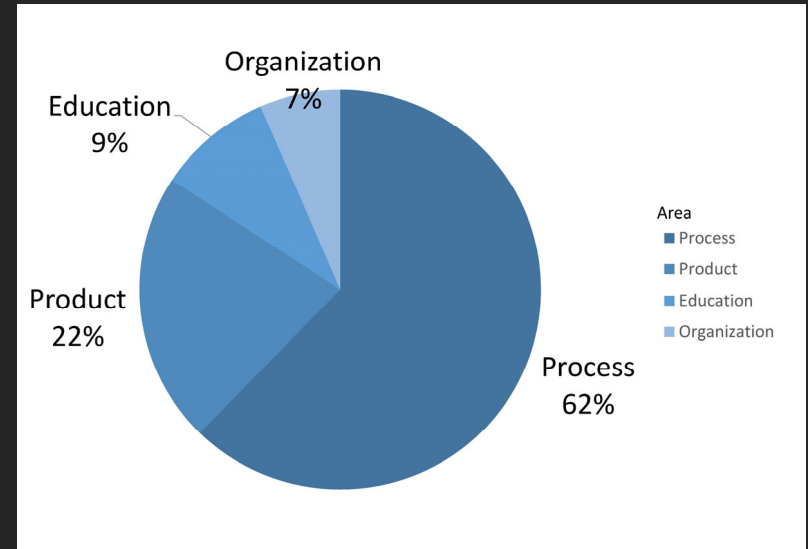
On average, the trend is stable (contrary to predictions by Dos Santos and Travassos (2011))



AR studies are mostly published in journals (IST, EMSE, JSS, RE)

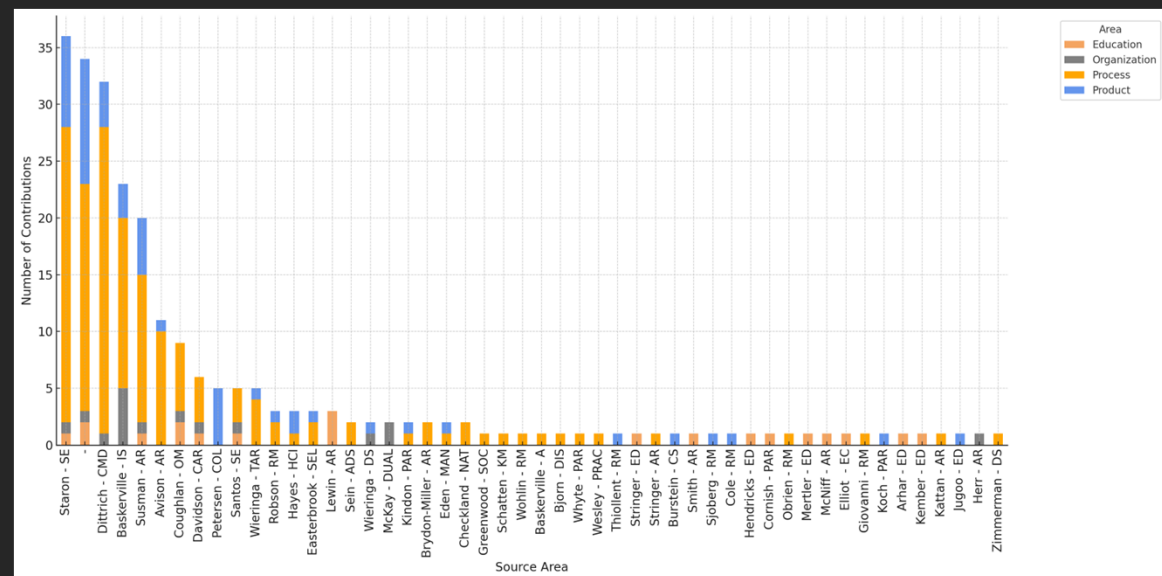
What AR studies improve

- Process
 - How to develop software?
- Product
 - How to improve the product?
- Education
 - How do SE professionals learn?
- Organization
 - How teams/companies operate?

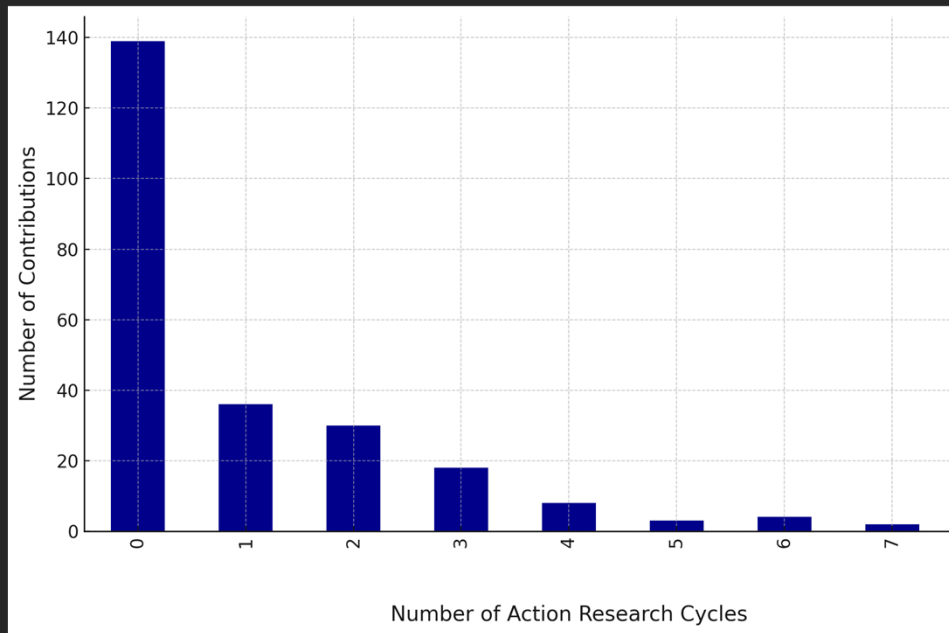


Which methodology is cited/used?

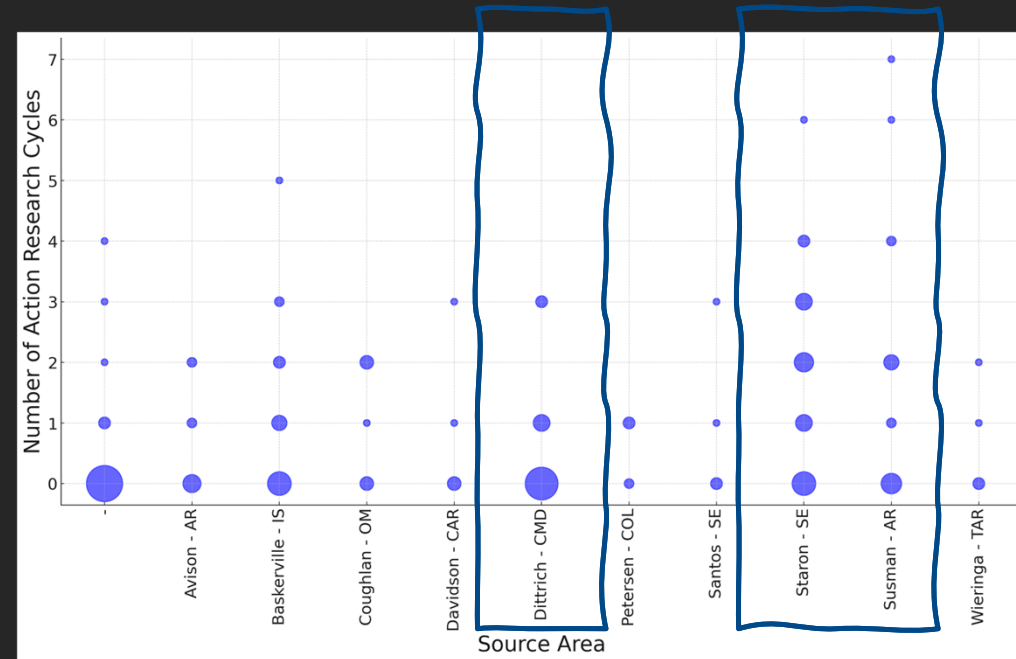
- Focus on the post-fix
 - SE – software engineering specific
 - CMD – method development specific
 - IS – Information systems
 - OM – Operations Management
 - CAR – Canonical Action Research
 - TAR – Technical Action Research (very close to DSR)



How many cycles are used?



The majority of studies do not report cycles (0), which could be either 1 cycle or many.



If a methodology prescribes cycles, then they are reported.

More cycles → longer papers → journals

Two example studies

Building a Learning Organization: Three Phases of Communities of Practice in a Software Consulting Company

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Torgeir Dingsøyr, Tore Dybå
SINTEF ICT, Norway
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Abstract

Special interest groups or communities of practice are seen as important mechanisms for knowledge sharing in the knowledge management literature. However, there are few empirical studies of communities of practice in the software engineering field. This article presents an action research project in a small size software consulting company, where the organization of the learning activities went through three phases, where different models were tried out and evaluated. We discuss the models in relation to community of practice theory to explain the learning effect of the models. A flexible model for organizing knowledge-sharing seems to provide the greatest learning effect.

Because of the complexity and extent of human judgment in software development, we think personalization strategies for software companies should be explored further. This paper presents a study of three models for sharing knowledge in a software consulting company through different types of interest groups. To discuss the three models, we use a theory to explain learning. We could have used various theories, like Nonaka and Takeuchi's theory of knowledge creation [13] or Argyris and Schön's theory of double-loop learning [1], but have decided to use the theory of Communities of Practice as this places special attention on the social elements of learning, which was a central element in the company under study.

The rest of the paper is organized as follows: First, we present general theory on communities of practice and then the research method applied in this study. Further

Data Sovereignty for AI Pipelines: Lessons Learned from an Industrial Project at Mondragon Corporation

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ABSTRACT

The establishment of collaborative AI pipelines, in which multiple organizations share their data and models, is often complicated by lengthy data governance processes and legal clarifications. Data sovereignty solutions, which ensure data is being used under agreed terms and conditions, are promising to overcome these problems. However, there is limited research on their applicability in AI pipelines. In this study, we extended an existing AI pipeline at Mondragon Corporation, in which sensor data is collected and subsequently forwarded to a data quality service provider with a data sovereignty component. By systematically reflecting and generalizing our experiences during the twelve-month action research project, we formulated ten lessons learned, four benefits, and three barriers to data-sovereign AI pipelines that can inform further research and custom implementations. Our results show that a data sovereignty component can help reduce existing barriers and increase the success of collaborative data science initiatives.

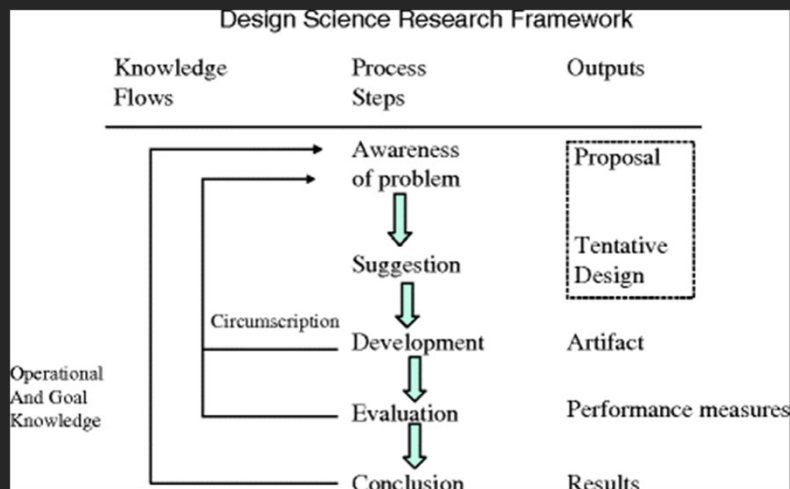
to reality [6, 27]. Vast amounts of available data have led to a democratization of AI and companies of different sizes and from different industries have started their own AI initiatives and experiments [10]. However, despite their broad adoption, AI and ML have not yet delivered on their promises in industrial practice [27]. The shift from prototypical projects to production-ready development is challenging, and many promising initiatives remain insular and isolated from other processes and systems [10, 27].

Since there is "no AI without data" [27] (p.98), data related challenges, including management, governance, and democratization, are the major obstacles for successful AI initiatives [5, 27]. They hinder organizations from leveraging potentially useful data sets and from fully exploiting the benefits of cooperatively working on joint AI workflows [27]. Hereby, the establishment of collaborative AI (e.g., a car manufacturer and its supplier working on a shared AI model) could help overcome insular initiatives and take full advantage of the benefits offered by AI.

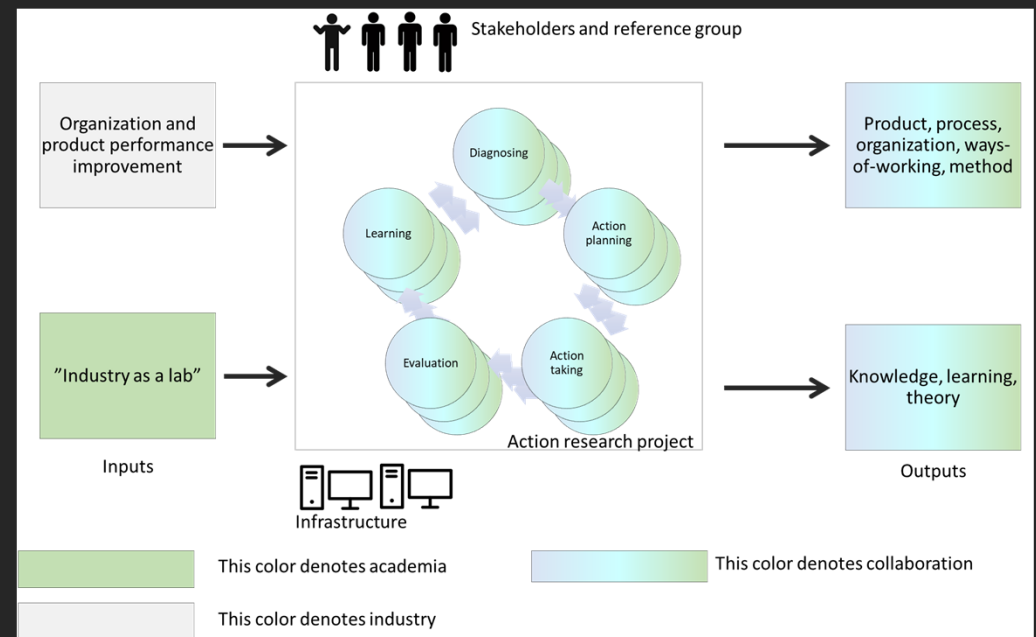
Mestad, Aase, et al. "Building a learning organization: Three phases of communities of practice in a software consulting company." *2007 40th Annual Hawaii International Conference on System Sciences (HICSS'07)*. IEEE, 2007.

Altendeitering, Marcel, et al. "Data sovereignty for AI pipelines: lessons learned from an industrial project at Mondragon corporation." *Proceedings of the 1st International Conference on AI Engineering: Software Engineering for AI*. 2022.

AR vs DSR – another view



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When to choose one over the other

	Action research	Design Science Research
Focus	Intervention and Action	Artifact and Treatment
Location	On-site	Off-site
Researchers	Industry or Industry + Academia	Academia or Academia + Industry
Contribution	Practice and theory	Practice and theory
Include learning	Yes, explicit	Yes, implicit in the evaluation
Evaluated entity	Action and its impact	Artifact and its design
Impact of research is primarily on	Client organization	Developed artifact
Intervention	In the problem setting of the client organization	In the local organization where the artifact is to be used

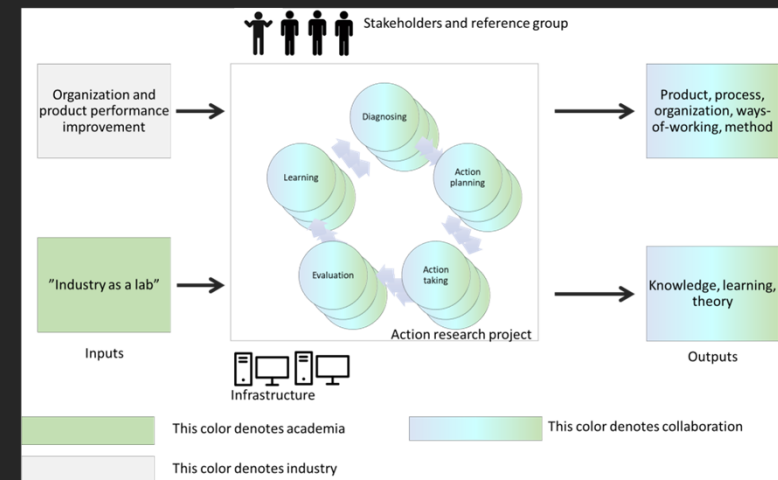
ETHICS IN AR

Areas of ethical considerations in AR

- Access, storage and processing of data
 - Completeness, security, safety, biases, ...
- Publication of results
 - Interpretation and under-/over- interpretation, bias in statistics, bias in opinions, claims,
- Contribution
 - To the host environment, to the scientific community, bias (political, organizational, market, ...)

Access, storage and processing of data

- Which/what data we are allowed to access
- How to access that data
 - On company premises (== company rules apply)
 - Take outside (== university rules apply + Swedish Public Information and Secrecy Act (2009:400))
- For what purpose
 - Which analysis we will do
- Unauthorized/unlawful access to data
 - Data outside of the scope of the project
 - Purpose/analysis outside of the project
 - Person outside of the project (e.g. manager)
 - Person unaware of the project (e.g. admin)



Data – data and meta-data example

Variable	Value
Average_transaction_time	6,33
number_of_customers	4

Analyzing data and finding
patterns, measurements,
KPIs, ...



Meta-data:

- About transactions
- Usually belongs to the agents/companies
- Usually we can analyze and sell it
- Unclear ownership (GDPR)
- Unclear responsibility

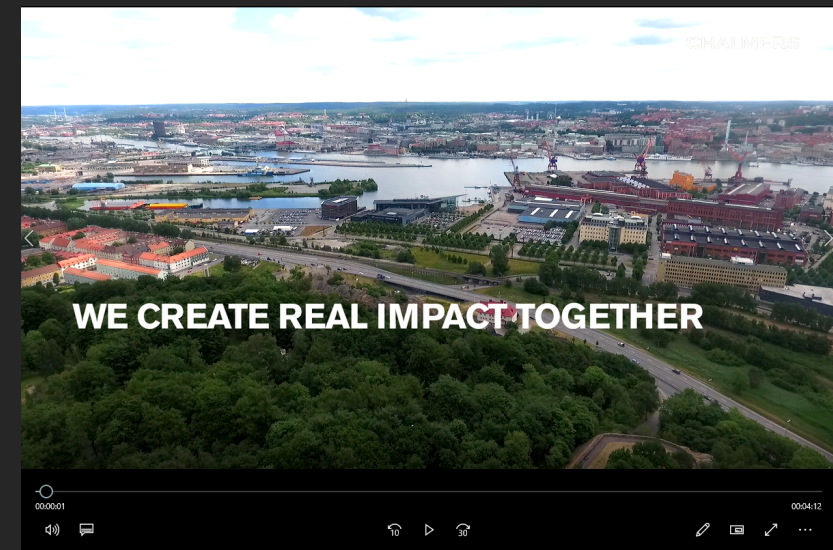
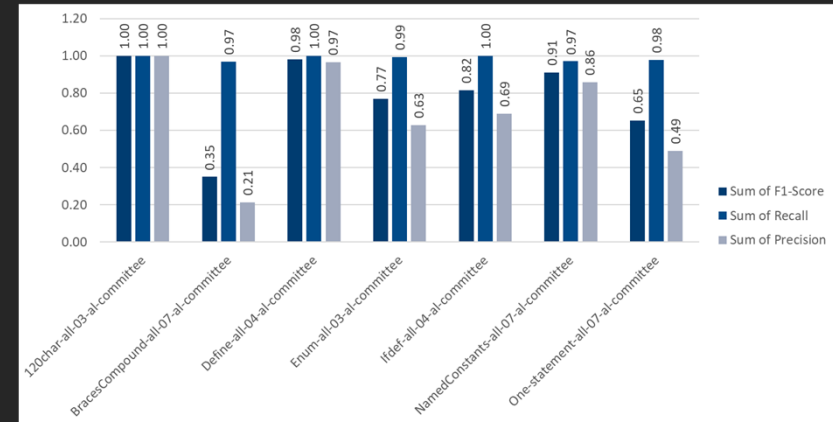
Customer	Transaction type	Price	Stock	start	end	status
John	Buy	10	APPL	2020-11-01	2020-11-02	OK
Mary	Sell	11	MSFT	2020-11-10	2020-11-20	Cancelled
Tom	Sell	11	MSFT	2020-12-02	2020-12-10	OK
Anna	Buy	23	GOGL	2020-12-13		Ongoing

Data:

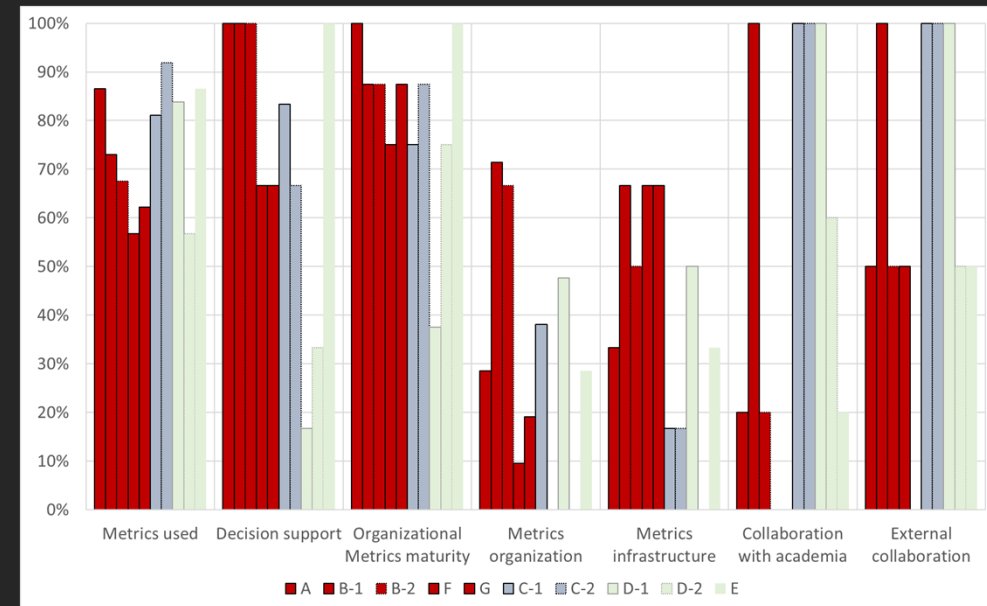
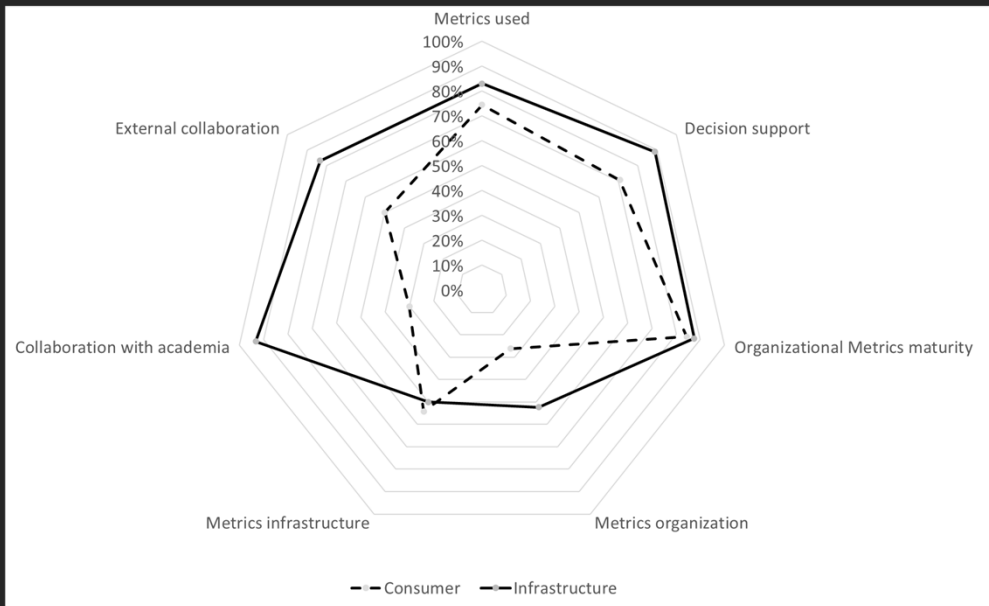
- Transactions
- Usually belongs to the customers
- Usually we can analyze it
- Usually we cannot sell it
- Clear ownership (GDPR)

Publication of results

- What we can and cannot publish (say)
- Process vs. Product vs. Organization
 - What we studied
 - What we have knowledge about
- Publish positive and negative results
 - Work with the company to find a good way to fulfill NDA obligations
- Increase company's involvement in publication from the very beginning
 - To learn what we need to publish, what we can publish and how to de-identify results
- Identify the right venue: industrial vs. scientific

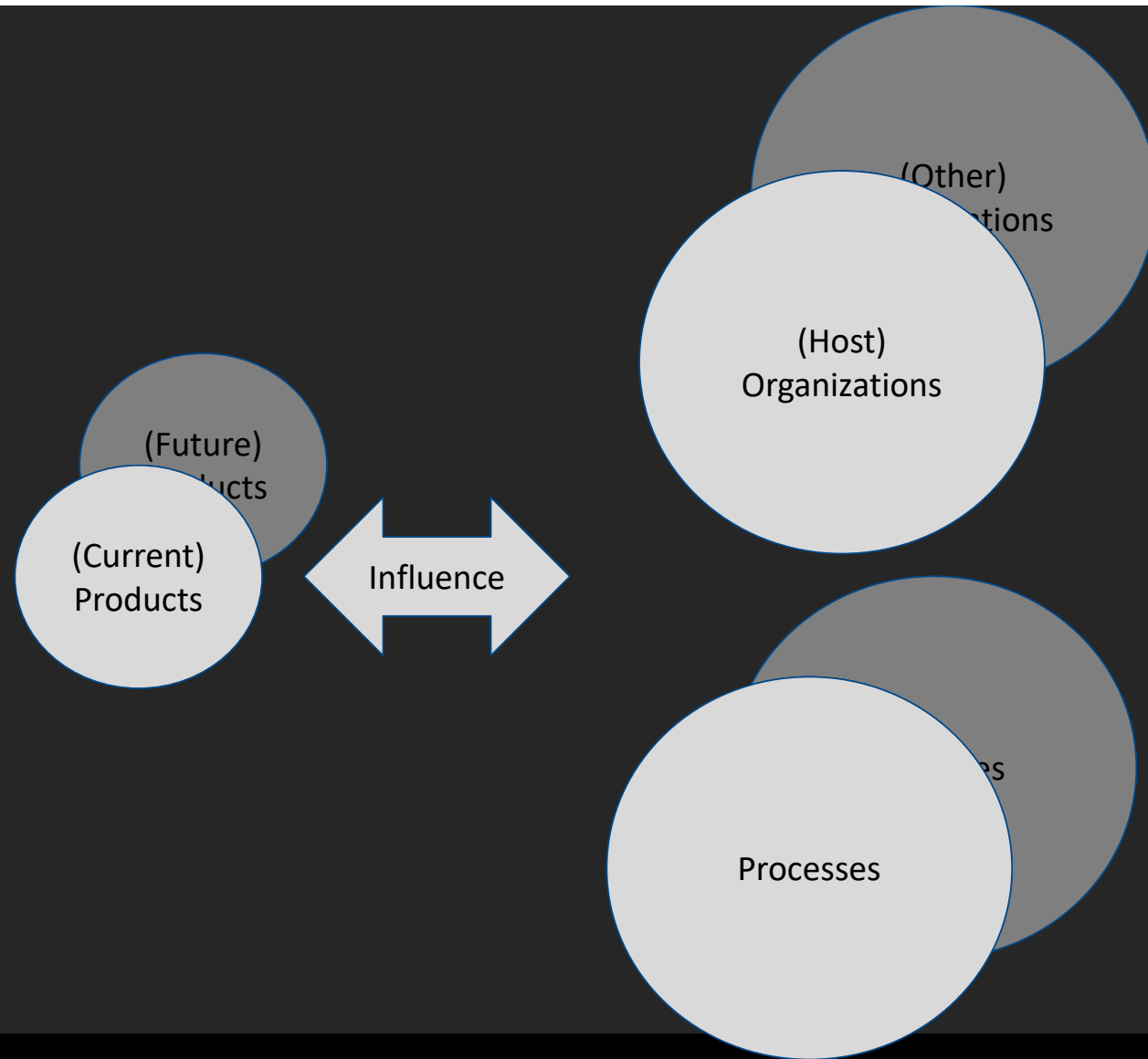


Example: Comparing maturity of companies



Contributions

- To the right element
 - AR is about action → processes and organizations are affected the most
- To the host processes
 - Improvements, new products, new processes
- To the scientific community
 - What we have learned



NDA, Contracts – protecting studies, companies and science

- Contracts
 - Used to protect the collaboration and all parties involved
 - Regulate who has the access to the information
 - Regulate what we can do with the information
 - Regulate what we can do within the company
 - Regulate the process to resolve disagreements
- NDA (Non-disclosure agreement)
 - Regulate the process of publishing
 - Regulate the process of accessing and processing data

Section 14 – Confidentiality

14.1 Confidentiality with regard to Parties who are Authorities or Equivalent to Authorities
In matters regarding Parties who are authorities, or who in matters of confidentiality shall be considered equivalent to authorities according to chapter 2, section 4 of the Swedish Public Access and Secrecy Act (2009:400), the regulations in said law shall apply.

According to this agreement, a Party does not have any obligation to provide Parties which are authorities, or which in matters of confidentiality shall be considered equivalent to authorities, with any Confidential Information, unless that authority or Party which in matters of confidentiality shall be considered equivalent to an authority, can classify the Confidential Information as secret according to the Swedish Public Access and Secrecy Act (2009:400). A Party may not provide Parties which are authorities, or which in matters of confidentiality shall be considered equivalent to authorities, with any Confidential Information which is owned by another Party, unless the owning Party has given prior written consent to such disclosure.

Parties which are authorities, or which in matters of confidentiality shall be considered equivalent to authorities, shall handle Background Information and unpublished Results with reasonable care, treat it confidential and ensure that such information is only provided to the persons, individuals or legal entities, which require the same for their work/assignments/operations according to this agreement.

Upon request, a Party shall indicate the persons at the Party which are authorized to receive Confidential Information from the other Party.

Jag, [namn och pers.nr], som är engagerad i Projektet vid Göteborgs universitet, är helt införstådd med villkoren i Avtalet, och accepterar härmed villkoren i Avtalet i för mig tillämpliga delar och förbinder mig att fullgöra mina skyldigheter i enlighet med Avtalet under den tidsperiod som Avtalet avser – även i det fall att jag väljer att avsluta min delaktighet i Projektet. Jag har särskilt uppmärksammat de villkor som rör immateriella rättigheter, publicering och sekretess och vilken påverkan detta kan få, dels för andra åtaganden som jag är bunden av, dels för de framtida åtaganden som jag kan tänkas vilja åta mig.

Jag upplåter härmed rätt till bakgrundsinformation samt upplåter eller överlåter eventuella resultat på det sätt och i den utsträckning som krävs för att Göteborgs universitet ska kunna fullgöra sina åtaganden enligt Avtalet.

NDAs, Contracts – cont...

- NDAs, e.g. [Mutual Non-Disclosure Agreement \(fccid.io\)](#)
 - Between company and a person
 - Between companies
- Regulates the liability for damages

1. **Definition.** “Confidential Information” shall mean confidential, non-public or other proprietary information including, without limitation, the existence of this Agreement, trade secrets, technical information, including algorithms, code, data, designs, documentation, drawings, formulae, hardware, know-how, ideas, inventions, whether patentable or not, photographs, plans, procedures, processes, reports, research, samples, sketches, software, specifications, business information, including customer and distributor names, marketing information, operations, plans, products, financial information, including pricing and other confidential information that is disclosed under the terms of this Agreement by Oceus or Company (“Disclosing Party”) to the other party (“Recipient”).

d) The Recipient of Confidential Information shall be permitted to disclose Confidential Information under this Section 2(d) and shall not be in violation of this Agreement if such Confidential Information is disclosed pursuant to the requirement or request of a duly empowered governmental agency or court of competent jurisdiction to the extent such disclosure is required by law, regulation or court order, and sufficient prior notice is given by the Recipient to the Disclosing Party of any such requirement or request to permit the Disclosing Party to seek an appropriate protective order to contest the disclosure of Confidential Information.

Each party recognizes that breach of this Agreement may cause irrevocable harm to the other party that would be inadequately compensable in damages and that the non-breaching party is entitled to seek injunctive relief for any such breach without proof of damages.

Table 9.3 Checklist for action planning—part II

Question	Comment for stakeholder	Comment for researcher
Is there a stakeholder appointed? Does the stakeholder have the mandate to drive and implement the results of the research project?	Make sure that (a) you are indeed the stakeholder of the research project and that (b) you have the mandate to implement the results of the research project in the company	As a researcher, you must have a counterpart in industry—a stakeholder. You may be able to write and submit a paper, but if the company does not have a “recipient” and/or does not gain anything from your research results, there will be able to perform more research projects again
	You must anchor these two questions with your company	Verify that the person you are working with has the mandate to drive and implement the results in the research project. Verify also that this is how the company understands it to be the case
Have security issues been addressed prior to the start of the research project?	You have the responsibility to go through security issues prior to the start of the research project. For example, • Has the researcher signed	Examples of security-related questions: • Have you signed the company’s NDA? • Can you take research-related

Table 9.4 (continued)

Question	Comment for stakeholder	Comment for researcher
During these regular meetings, are the work-related issues discussed/followed up on?	During the stay of the researchers from the action team, you have the responsibility for him/her, including his/her working environment. For example, that the researcher has a good working place (desk and computer access), access to related data, and sitting close to you and/or by the employees he/she will work with	Though you are guest at the company, you should have the same working environment standard as its employees. Do not accept anything less! When it comes to working in the company, respect that the employees (most often) have much to do and may not always be available for you. It is better to have more shortly meetings that a few lengthy ones. A tip is to sit close to those you are working with!

SUMMARY

Action Research in Software Engineering

- Cornerstones
 - Intervention
 - Host organization
- Cycles
 - diagnose -> plan -> do -> evaluate -> learn -> diagnose -> ...
- Not for everyone
 - AR vs. Design Science vs. Case Studies vs. Experiments
- Where to go from here
 - Plan your thesis based on AR – if you have the two cornerstones
 - Use AR in your future career – learn and study as much as do and deliver

